The Norm of $P_n(x)$, Method 2: Via the Generating Function

- For $m = n$, write the generating function twice, once summed over n and once over m (both in the same variable t):

$$(1-2xt+t^2)^{-1/2} = \sum_n P_n(x)t^n, \qquad (1-2xt+t^2)^{-1/2} = \sum_m P_m(x)t^m.$$

- Multiplying and integrating over $[-1, 1]$, all cross terms ($m \neq n$) vanish by orthogonality, leaving

$$\int_{-1}^{1} (1 - 2xt + t^2)^{-1} dx = \sum_{n=0}^{\infty} t^{2n} \int_{-1}^{1} [P_n(x)]^2 \, dx.$$

- For $t = 0$, the LHS is $\int_{-1}^{1} dx = 2$.
- For $t = \pm 1$, the integral diverges to ∞.



Method 2: Completing the Calculation

- For $|t| < 1$, the integration of the left side gives

$$\int_{-1}^{1} \frac{dx}{1-2xt+t^2} = \frac{1}{t} \ln \frac{1+t}{1-t} = \frac{2}{t}\left[t + \frac{t^3}{3} + \frac{t^5}{5} + \cdots\right] = \sum_{n=0}^{\infty} \frac{2t^{2n}}{2n+1}.$$

- Comparing coefficients of t^{2n} on both sides:

$$\int_{-1}^{1} [P_n(x)]^2 \, dx = \frac{2}{2n+1}.$$

- Together with the $m \neq n$ result, this proves both (I) and (II).



Orthogonality Relation: Summary

- Combining both results:

$$\int_{-1}^{1} P_m(x)P_n(x)\,dx = \begin{cases} 0 & m \neq n, \\ \dfrac{2}{2n+1} & m = n. \end{cases}$$

- So $\{P_n(x)\}$ is an orthogonal (but not orthonormal) family on $[-1, 1]$ with weight $w(x) = 1$.
- The corresponding orthonormal family is $\left\{ \sqrt{\frac{2n+1}{2}}\, P_n(x) \right\}$.



Expanding Polynomials in Legendre Series

- Many engineering problems require expanding a given function in a series of Legendre polynomials.
- Recall $P_0 = 1$, $P_1 = x$, $P_2 = \frac{1}{2}(3x^2 - 1)$, $P_3 = \frac{1}{2}(5x^3 - 3x)$.
- Inverting these relations, any cubic can be rewritten as

$$b_0 + b_1 x + b_2 x^2 + b_3 x^3 = \left(b_0 + \frac{b_2}{3}\right) P_0(x) + \left(b_1 + \frac{3b_3}{5}\right) P_1(x)$$

$$+ \frac{2b_2}{3} P_2(x) + \frac{2b_3}{5} P_3(x).$$

- Any polynomial of degree n can similarly be expanded in $P_0(x), \ldots, P_n(x)$ — this is always a *finite*, exact expansion.



Problem : Legendre Series for $f(x) = x^2$

- Expand $f(x) = x^2$ in terms of Legendre polynomials.
- Since f is already a polynomial of degree 2, we need only P_0, P_1, P_2.
- Write $x^2 = a_0 P_0(x) + a_2 P_2(x) = a_0 + a_2 \cdot \frac{1}{2}(3x^2 - 1)$ (the P_1 coefficient is 0 since x^2 is even).
- Matching coefficients: $\frac{3}{2}a_2 = 1 \Rightarrow a_2 = \frac{2}{3}$, and $a_0 - \frac{1}{2}a_2 = 0 \Rightarrow a_0 = \frac{1}{3}$.
- So

$$x^2 = \frac{1}{3}P_0(x) + \frac{2}{3}P_2(x).$$

- This is an *exact*, finite expansion — no infinite tail, since x^2 is itself a polynomial of degree 2.



Problem : Legendre Series for $f(x) = 2x^3 - x + 1$

- Expand $f(x) = 2x^3 - x + 1$ using the cubic expansion formula with $b_0 = 1,\ b_1 = -1,\ b_2 = 0,\ b_3 = 2$:

$$f(x) = \left(b_0 + \frac{b_2}{3}\right) P_0 + \left(b_1 + \frac{3b_3}{5}\right) P_1 + \frac{2b_2}{3} P_2 + \frac{2b_3}{5} P_3.$$

- Substituting: coefficient of P_0 is $1 + 0 = 1$; of P_1 is $-1 + \frac{6}{5} = \frac{1}{5}$; of P_2 is 0; of P_3 is $\frac{4}{5}$.
- So

$$2x^3 - x + 1 = P_0(x) + \frac{1}{5} P_1(x) + \frac{4}{5} P_3(x).$$

- Check:
$1 + \frac{1}{5}x + \frac{4}{5} \cdot \frac{1}{2}(5x^3 - 3x) = 1 + \frac{x}{5} + 2x^3 - \frac{6x}{5} = 1 - x + 2x^3.$



The Vector Space of Polynomials, Revisited

- Recall $\mathcal{P}(x)$, the vector space of all polynomials, has the standard (infinite) basis $\{1, x, x^2, \dots\}$.
- Since each $P_n(x)$ has degree exactly n, the set $\{P_0(x), P_1(x), P_2(x), \dots\}$ is *also* a basis of $\mathcal{P}(x)$ — and, unlike the standard one, an **orthogonal** basis.
- This is exactly why every polynomial $f(x)$ admits a finite, exact expansion in Legendre polynomials: it is simply a change of basis within $\mathcal{P}(x)$, as seen in the previous example.
- For general (non-polynomial) f, the analogous statement requires the infinite-dimensional setting of $L^2([-1, 1])$ — this is the subject of the Fourier–Legendre series, developed next.



The General Legendre Series

- If $f(x)$ for $-1 < x < 1$ can be written as $f(x) = \sum_{n=0}^{\infty} a_n P_n(x)$, we say f has a **Legendre series expansion**.
- To find the a_n: multiply by $P_m(x)$, integrate term by term, and use orthogonality:

$$\int_{-1}^{1} f(x)P_m(x)\,dx = \sum_{n=0}^{\infty} a_n \int_{-1}^{1} P_m(x)P_n(x)\,dx = \frac{2a_m}{2m+1}.$$

- Therefore

$$\boxed{a_n = \left(n + \frac{1}{2}\right) \int_{-1}^{1} f(x)P_n(x)\,dx}$$



Example: Fourier–Legendre Series

- Find the first 4 non-zero terms of the Fourier–Legendre series for

$$f(x) = \begin{cases} x & 0 \le x \le 1, \\ 0 & -1 \le x \le 0. \end{cases}$$

- Here $a_n = \dfrac{2n+1}{2} \displaystyle\int_0^1 x P_n(x)\, dx$, $n \ge 0$:

$$a_0 = \frac{1}{2}\int_0^1 x\, dx = \frac{1}{4}, \qquad a_1 = \frac{3}{2}\int_0^1 x \cdot x\, dx = \frac{1}{2},$$

$$a_2 = \frac{5}{2}\int_0^1 x \cdot \frac{3x^2-1}{2}\, dx = \frac{5}{8}, \qquad a_3 = \frac{7}{2}\int_0^1 x \cdot \frac{5x^3-3x}{2}\, dx = -\frac{7}{16}.$$

- So $f(x) = \frac{1}{4}P_0(x) + \frac{1}{2}P_1(x) + \frac{5}{8}P_2(x) - \frac{7}{16}P_3(x) + \cdots$, converging to $f(x)$ for all $x \in [-1, 1]$.



Problem : Fourier–Legendre Series for $f(x) = |x|$

- Find the first three non-zero terms of the Legendre series for $f(x) = |x|$ on $[-1, 1]$.
- Since f is even, $a_{2n+1} = 0$ for all $n \geq 0$; we need only even-indexed coefficients.

$$a_0 = \frac{1}{2}\int_{-1}^{1} |x|\,dx = \int_{0}^{1} x\,dx = \frac{1}{2}.$$

$$a_2 = \frac{5}{2}\int_{-1}^{1} |x|P_2(x)\,dx = 5\int_{0}^{1} x \cdot \frac{3x^2-1}{2}\,dx = \frac{5}{8}.$$

$$a_4 = \frac{9}{2}\int_{-1}^{1} |x|P_4(x)\,dx = 9\int_{0}^{1} x \cdot \frac{35x^4-30x^2+3}{8}\,dx = -\frac{3}{16}.$$

- So

$$|x| \sim \frac{1}{2}P_0(x) + \frac{5}{8}P_2(x) - \frac{3}{16}P_4(x) + \cdots$$

- Since $|x|$ is continuous on $(-1, 1)$, the series converges to $|x|$ everywhere on $[-1, 1]$.



Expressing a Quartic in Legendre Polynomials

- Express $f(x) = x^4 + 2x^3 + 2x^2 - x - 3$ in terms of Legendre polynomials.
- Invert the definitions of $P_0, \ldots, P_4$ to write $x^4, x^3, x^2, x, 1$ each in terms of $P_0, \ldots, P_4$, then substitute in succession (highest power first) and collect like terms.
- The result is

$$f(x) = -\frac{32}{15}P_0(x) + \frac{1}{5}P_1(x) + \frac{40}{21}P_2(x) + \frac{4}{5}P_3(x) + \frac{8}{35}P_4(x).$$

- One can use orthogonality to compute a_n, $n = 0, 1, 2, 3, 4$.
- Note that $a_n = 0$ for $n \geq 5$.
- Check (constant term at $x = 0$): substituting $x = 0$ into both sides and using $P_{2k+1}(0) = 0$ confirms consistency with $f(0) = -3$.



Fourier–Legendre Series and Square-Integrable Functions

- $f(x)$ on $[-1,1]$ is **square-integrable** if $\displaystyle\int_{-1}^{1} f(x)^2\,dx < \infty$.
- Polynomials, continuous functions, and piecewise continuous functions are all square-integrable.
- The set of all square-integrable functions on $[-1,1]$ is a vector space, denoted $L^2([-1,1])$.
- With weight $w(x) = 1$, define the inner product
  $\langle f, g\rangle = \displaystyle\int_{-1}^{1} f(x)g(x)\,dx.$



Legendre Polynomials as a Maximal Orthogonal Set

- The Legendre polynomials do *not* form a basis for $L^2([-1,1])$ in the finite-dimensional sense.
- But they form a **maximal orthogonal set** in $L^2([-1,1])$: there is no nonzero square-integrable function orthogonal to *every* $P_n(x)$.
- This maximality is exactly what makes the Legendre series expansion of an arbitrary $f \in L^2([-1,1])$ possible.



Convergence in the L^2 Norm

- The Fourier–Legendre series of $f \in L^2([-1,1])$,

$$\sum_{n=0}^{\infty} a_n P_n(x), \qquad a_n = \frac{2n+1}{2} \int_{-1}^{1} f(x) P_n(x)\, dx,$$

converges in L^2-**norm**: $\left\| f - \sum_{n=0}^{m} a_n P_n \right\|_2 \to 0$ as $m \to \infty$.

- Pointwise convergence is a more delicate matter, addressed next.



Pointwise Convergence: The Legendre Expansion Theorem

- Two natural questions for pointwise convergence: (i) does the series converge at a given x? (ii) if so, does it converge to $f(x)$?
- **Legendre Expansion Theorem:** If both $f(x)$ and $f'(x)$ have at most finitely many jump discontinuities on $[-1, 1]$, then the Legendre series converges to

$$\tfrac{1}{2}\left[f(x^+) + f(x^-)\right] \quad \text{for every } x \in (-1, 1),$$

  to $f(-1^+)$ at $x = -1$, and to $f(1^-)$ at $x = 1$.
- In particular, the series converges to $f(x)$ at every point of continuity.
- This mirrors the classical Fourier series convergence theorem exactly.



Example: Fourier–Legendre Series

- Find the first three non-zero terms for

$$f(x) = \begin{cases} 1 & 0 < x < 1, \\ -1 & -1 < x < 0, \end{cases}$$

and discuss convergence at $x = 0$.

- Since f is **odd** and P_{2n} is even, $a_{2n} = 0$ for all $n \geq 0$.

$$a_1 = \frac{3}{2}\int_{-1}^{1} f(x)\, x\, dx = 3\int_{0}^{1} x\, dx = \frac{3}{2},$$

$$a_3 = \frac{7}{2}\int_{-1}^{1} f(x)\cdot\frac{5x^3 - 3x}{2}\, dx = -\frac{7}{8},$$

$$a_5 = \frac{11}{2}\int_{-1}^{1} f(x)P_5(x)\, dx = 11\int_{0}^{1} f(x)P_5(x)\, dx = \frac{11}{16}.$$

- So $f(x) \sim \frac{3}{2}P_1(x) - \frac{7}{8}P_3(x) + \frac{11}{16}P_5(x) + \cdots$
- By the expansion theorem, the series converges to $f(x)$ if $x \neq 0$, and to $\frac{1}{2}[f(0^+) + f(0^-)] = 0$ at $x = 0$ — the midpoint of the jump.



Least-Squares Polynomial Approximation

- One more property sets the Legendre series apart: its partial sums are not just convergent — they are **optimal** polynomial approximations.
- **Theorem.** Let $f \in L^2([-1,1])$, and suppose we approximate f by a polynomial $p(x)$ of degree $\leq n$ so as to minimize

$$I = \int_{-1}^{1} \big[f(x) - p(x)\big]^2 \, dx.$$

- Then the minimizing polynomial is *exactly* the $(n+1)$-term truncation of the Legendre series of f:

$$p(x) = c_0 P_0(x) + \cdots + c_n P_n(x), \qquad c_k = \frac{2k+1}{2} \int_{-1}^{1} f(x) P_k(x) \, dx.$$

- **Caution:** if f has a power (Taylor) series on $[-1,1]$, the best least-squares polynomial approximation is generally *not* the same as truncating that power series.



Least-Squares Approximation: Proof

- Write a general degree-$\leq n$ polynomial as $p(x) = \sum_{k=0}^{n} b_k P_k(x)$. Expanding I and using orthogonality to kill all cross terms $P_j P_k$ $(j \neq k)$:

$$I = \int_{-1}^{1} f(x)^2 dx + \sum_{k=0}^{n} \frac{2}{2k+1} b_k^2 - 2 \sum_{k=0}^{n} b_k \int_{-1}^{1} f(x) P_k(x)\, dx.$$

- Since $\int_{-1}^{1} f P_k\, dx = \dfrac{2c_k}{2k+1}$, this becomes, after completing the square in each b_k,

$$I = \int_{-1}^{1} f^2\, dx + \sum_{k=0}^{n} \frac{2}{2k+1} (b_k - c_k)^2 \; - \; \sum_{k=0}^{n} \frac{2}{2k+1} c_k^2.$$

- The only b_k-dependent term is $\sum_k \frac{2}{2k+1}(b_k - c_k)^2 \geq 0$, minimized (at 0) exactly when $b_k = c_k$ for every $k = 0, \ldots, n$.
- So the least-squares polynomial is precisely the Legendre-series truncation. ■

